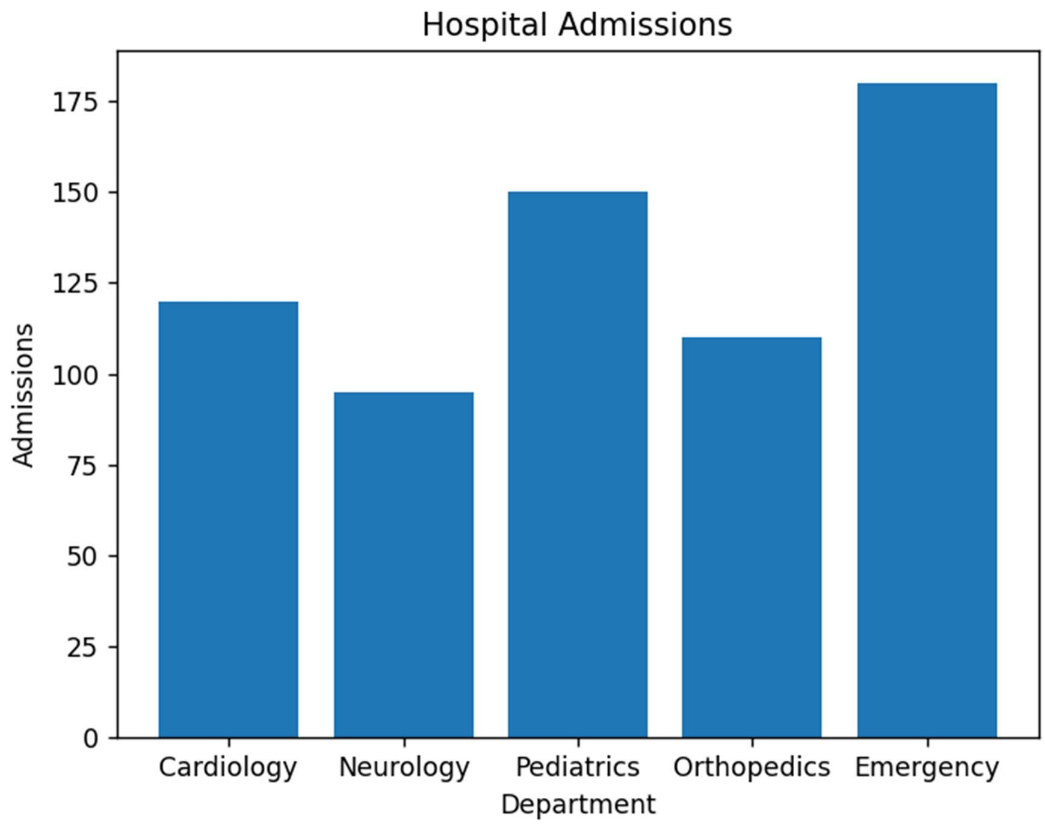
Hospital Patient Admission Analysis

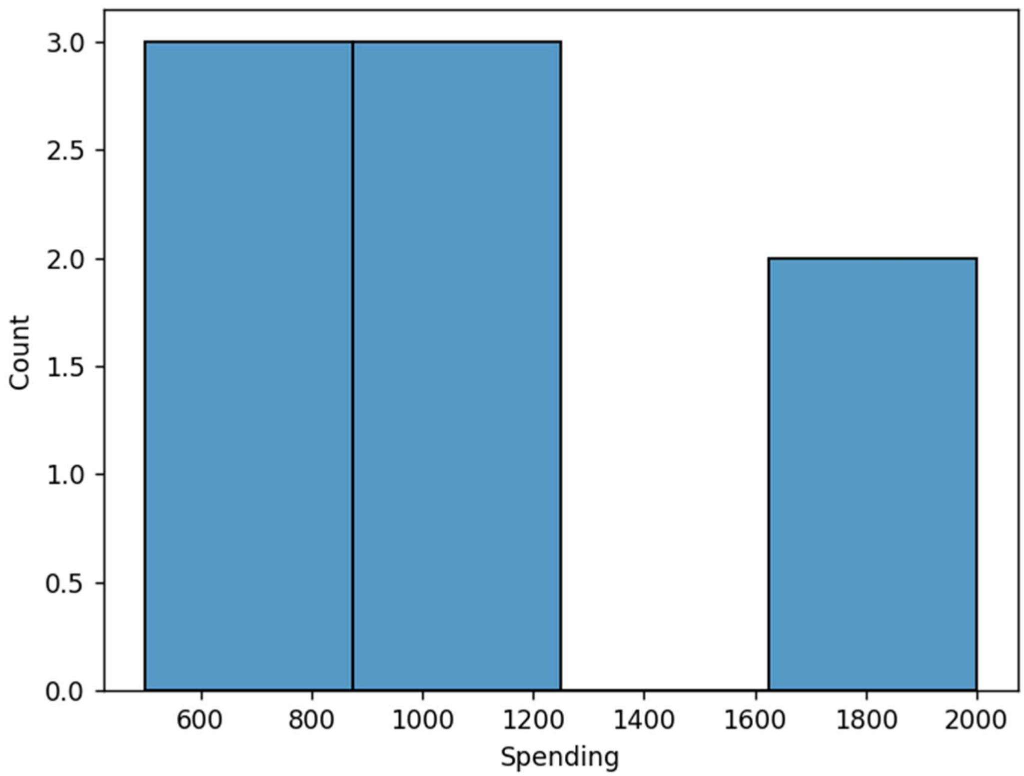
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = {
    "Department":["Cardiology","Neurology","Pediatrics","Orthopedics","Emergency"],
    "Admissions":[120,95,150,110,180]
}

df = pd.DataFrame(data)

plt.bar(df["Department"], df["Admissions"])
plt.title("Hospital Admissions")
plt.xlabel("Department")
plt.ylabel("Admissions")
plt.show()

sns.boxplot(y=df["Admissions"])
plt.show()

sns.histplot(df["Admissions"], bins=5)
plt.show()
```



E-Commerce Customer Spending Behaviour

```
import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt


df = pd.DataFrame({

    "Spending":[500,700,800,1200,2000,1800,900,1000]

})


sns.histplot(df["Spending"])

plt.show()


sns.violinplot(y=df["Spending"])

plt.show()
```

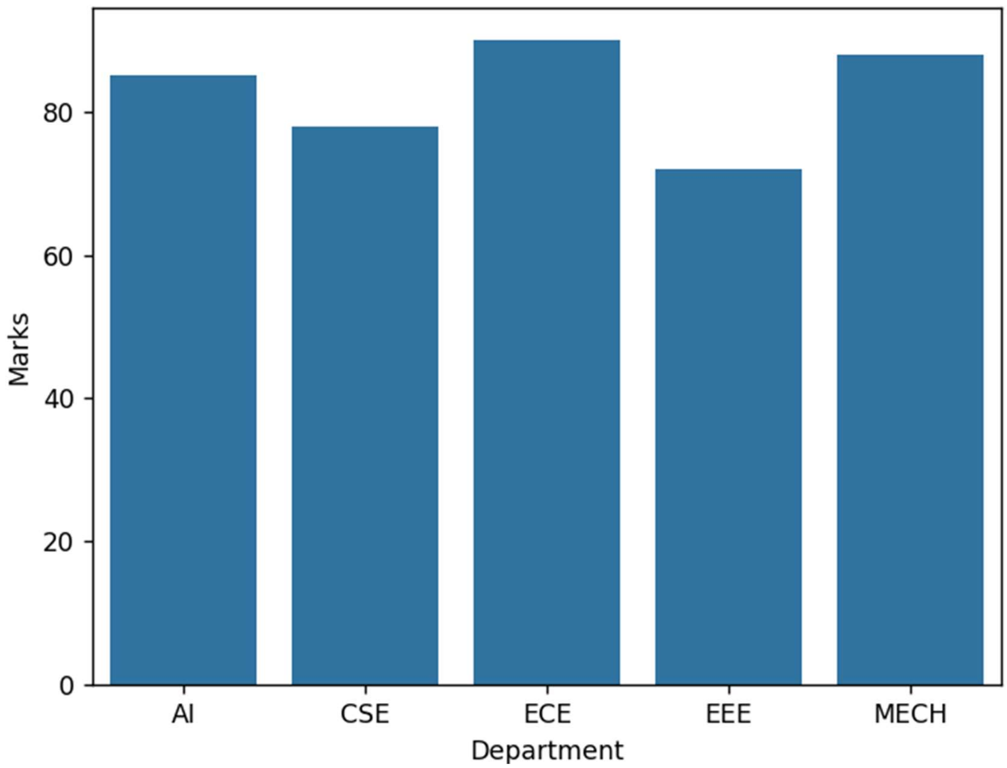
```
sns.boxplot(y=df["Spending"])
plt.show()

sns.kdeplot(df["Spending"], fill=True)
plt.show()
```



University Examination Performance Dashboard

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Department":["AI","CSE","ECE","EEE","MECH"],
    "Marks":[85,78,90,72,88]
})

sns.barplot(x="Department", y="Marks", data=df)
```

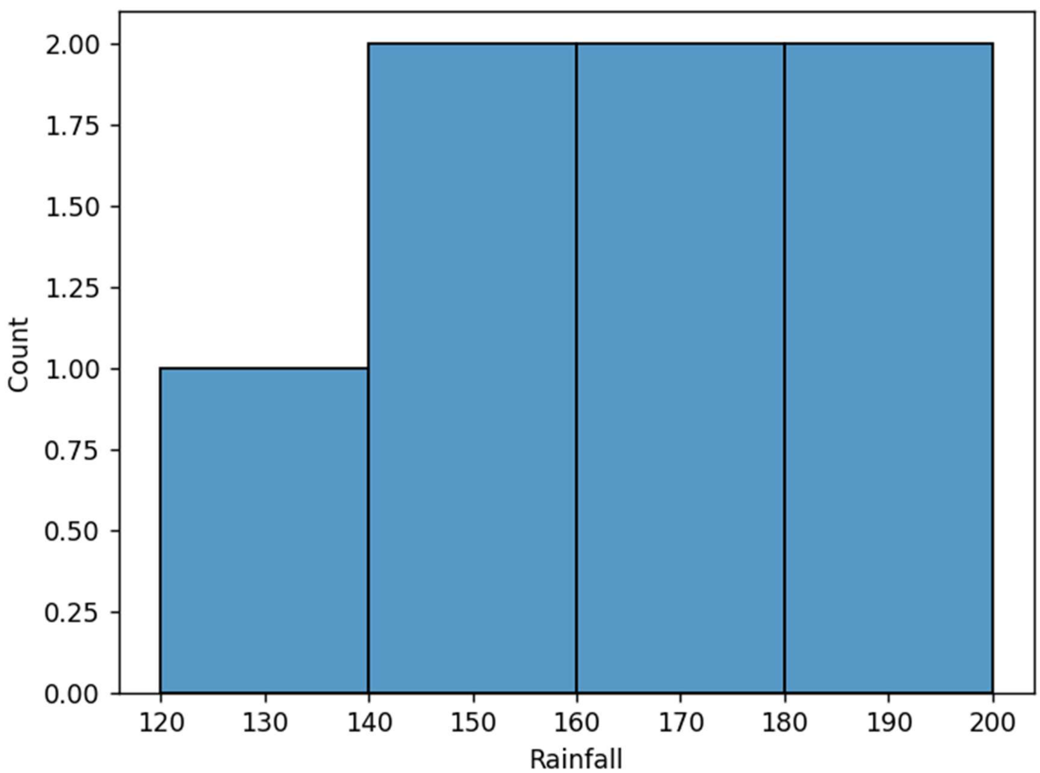
```
plt.show()

sns.boxplot(y=df["Marks"])
plt.show()

sns.histplot(df["Marks"])
plt.show()
```



Climate and Rainfall Distribution Study

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Rainfall":[120,150,180,200,160,170,140],
    "Temperature":[28,30,31,29,27,26,25]
})
```

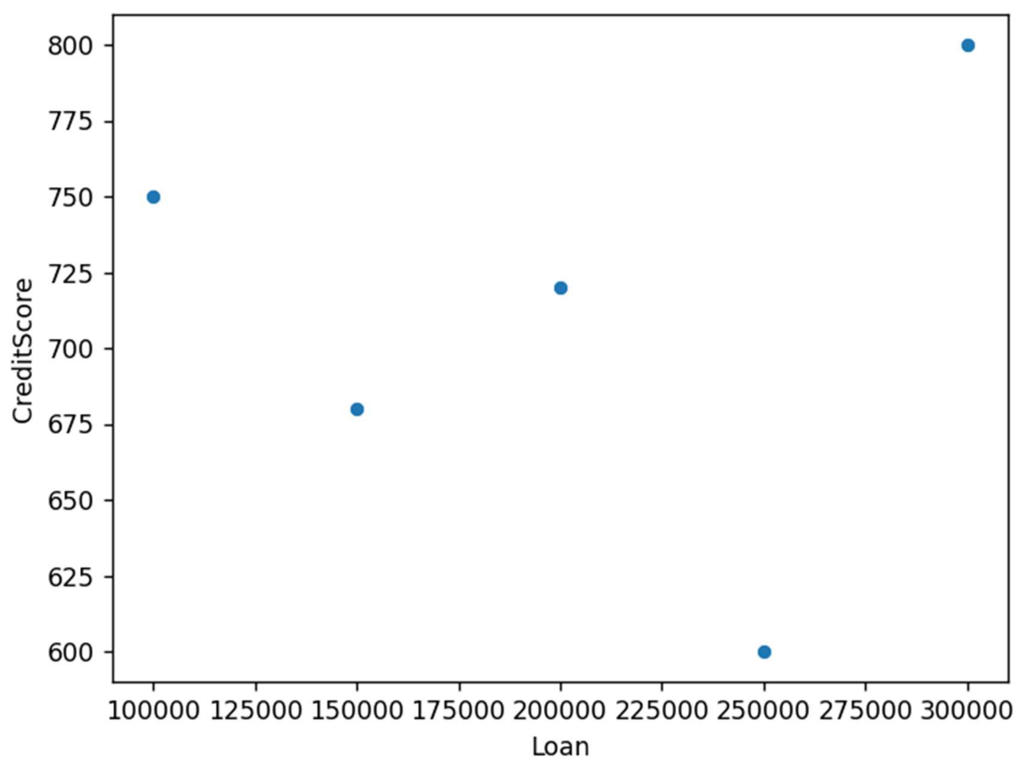
```
sns.histplot(df["Rainfall"])
plt.show()

sns.boxplot(y=df["Rainfall"])
plt.show()

sns.kdeplot(df["Rainfall"], fill=True)
plt.show()
```



Bank Loan Risk Assessment

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Loan":[100000,150000,200000,250000,300000],
```

```
    "CreditScore":[750,680,720,600,800]
})

sns.scatterplot(x="Loan", y="CreditScore", data=df)
plt.show()

sns.boxplot(y=df["Loan"])
plt.show()

sns.histplot(df["Loan"])
plt.show()
```

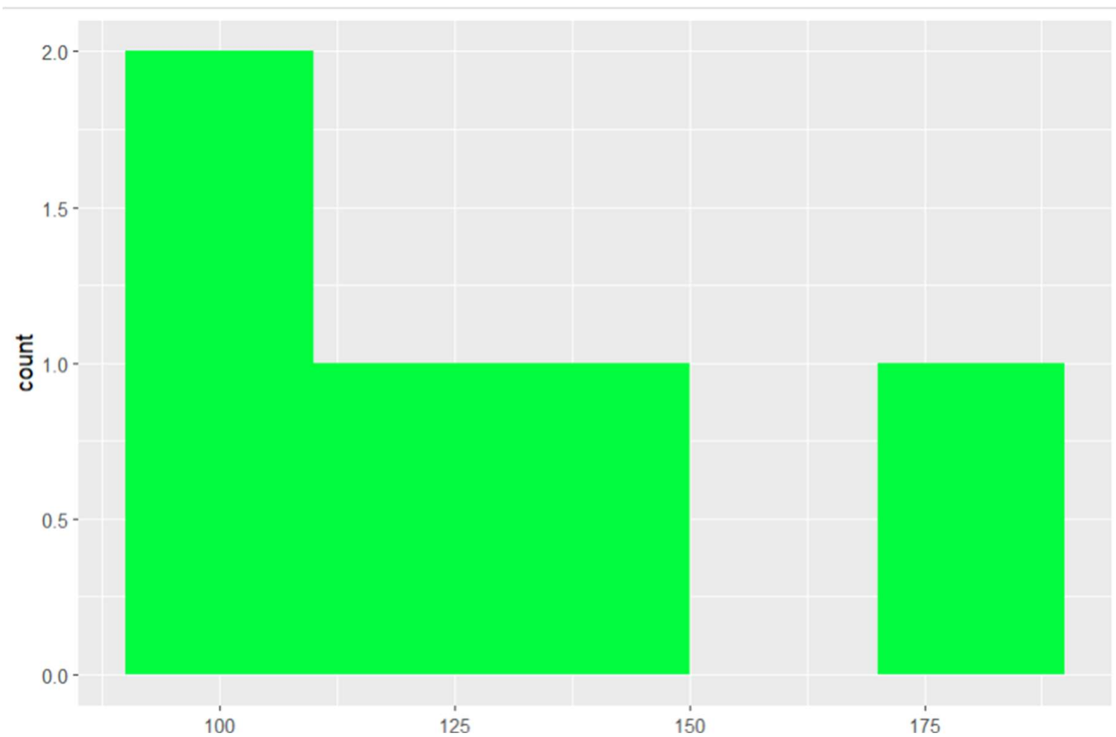


R programming

library(ggplot2)

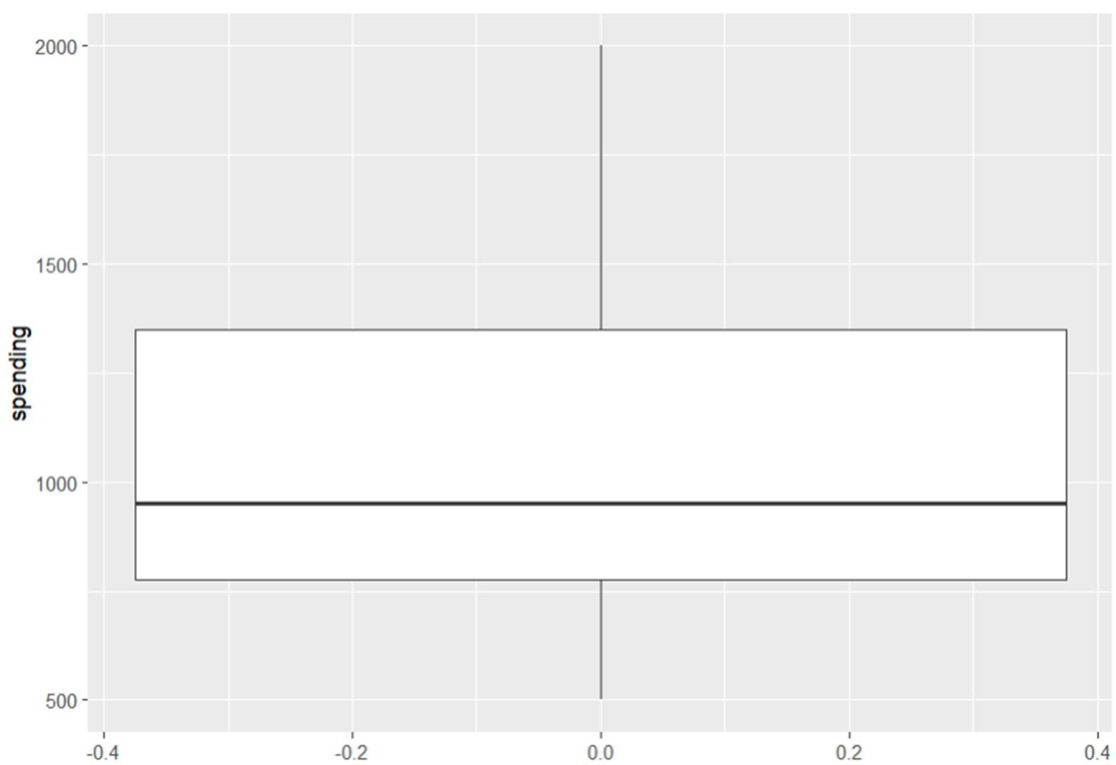
```
dept <- c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency")
admissions <- c(120,95,150,110,180)
```

```
df <- data.frame(dept, admissions)

ggplot(df, aes(dept, admissions)) +
geom_bar(stat="identity", fill="skyblue")

ggplot(df, aes(y=admissions)) +
geom_boxplot()

ggplot(df, aes(admissions)) +
geom_histogram(binwidth=20, fill="green")
```



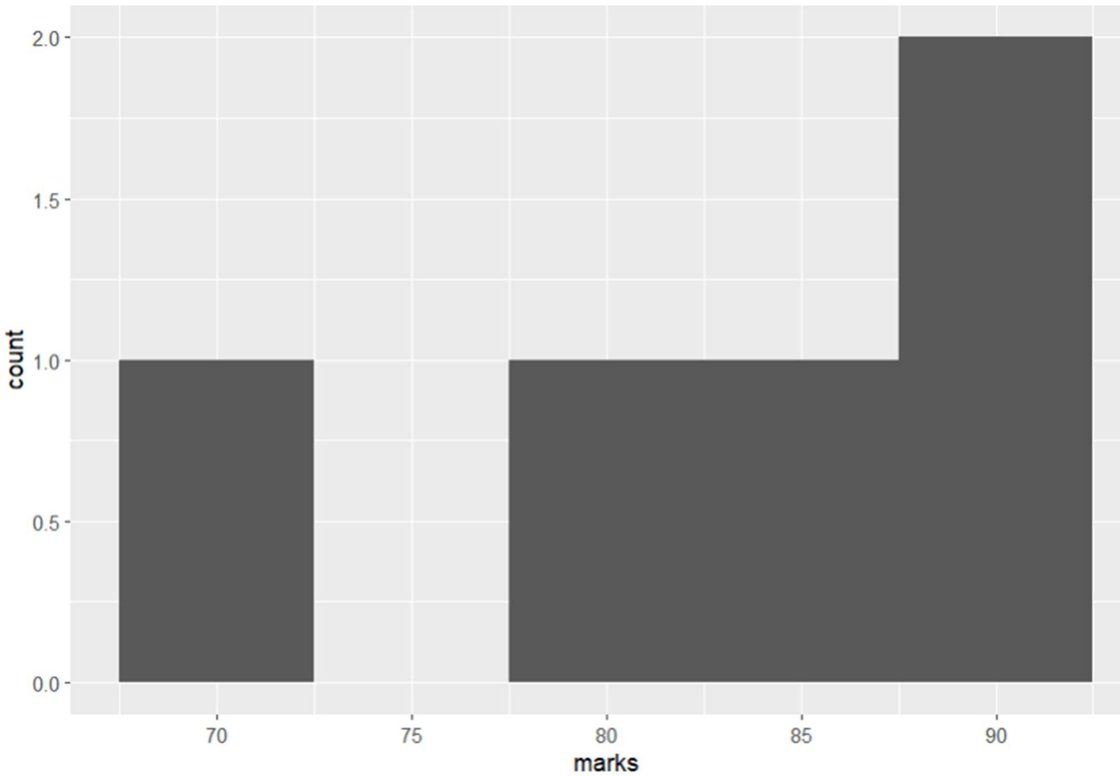

```
library(ggplot2)

spending <- c(500,700,800,1200,2000,1800,900,1000)
df <- data.frame(spending)

ggplot(df,aes(spending))+
```

```
geom_histogram(fill="orange")

ggplot(df,aes(y=spending,x=""))+
geom_violin(fill="pink")

ggplot(df,aes(y=spending))+
geom_boxplot()
```

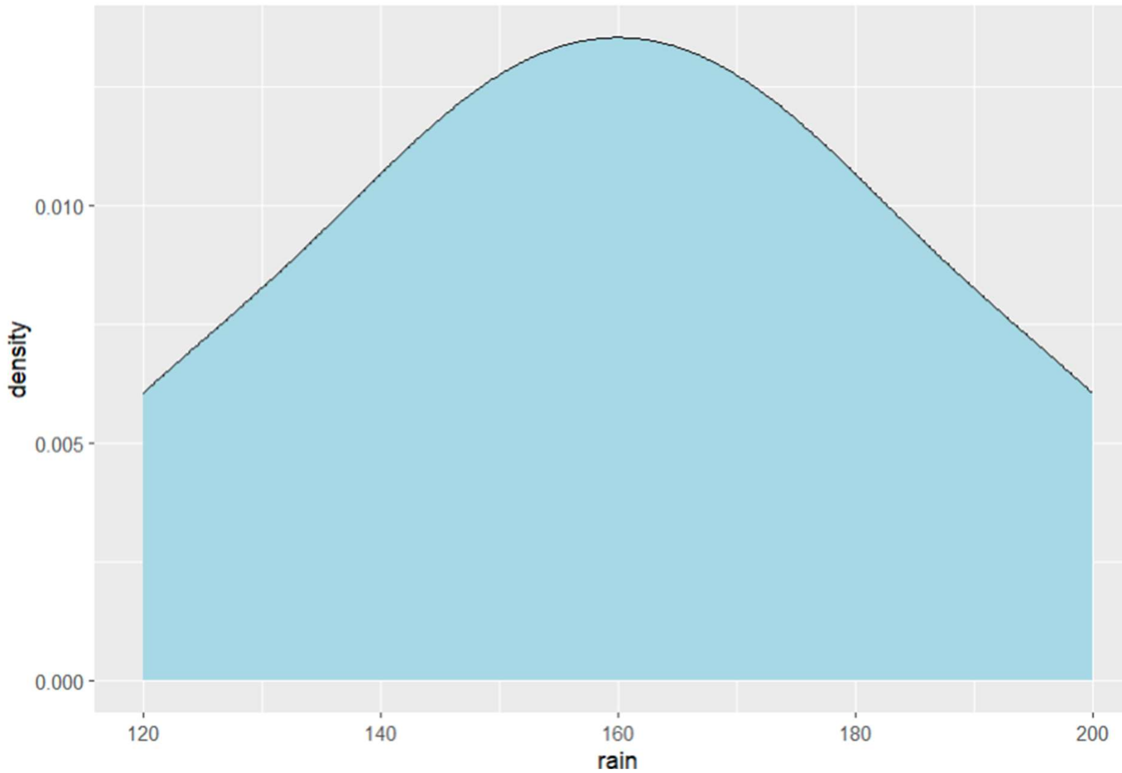



```
library(ggplot2)

dept <- c("AI","CSE","ECE","EEE","MECH")
marks <- c(85,78,90,72,88)

df <- data.frame(dept,marks)

ggplot(df,aes(dept,marks))+
geom_bar(stat="identity")

ggplot(df,aes(y=marks))+
```

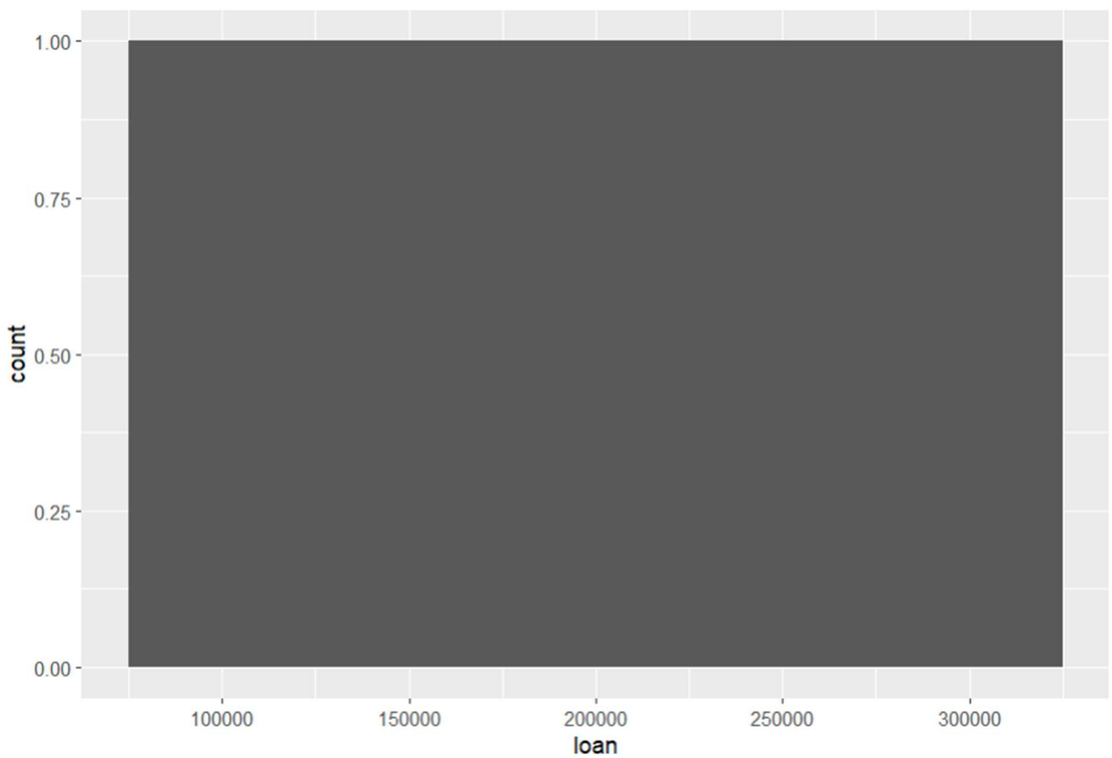
```
geom_boxplot()

ggplot(df,aes(marks))+
geom_histogram(binwidth=5)
\
```




```
library(ggplot2)

rain <- c(120,150,180,200,160,170,140)
df <- data.frame(rain)

ggplot(df,aes(rain))+
geom_histogram(fill="blue")

ggplot(df,aes(y=rain))+
geom_boxplot()

ggplot(df,aes(rain))+
geom_density(fill="lightblue")
```




```
library(ggplot2)

loan <- c(100000,150000,200000,250000,300000)
score <- c(750,680,720,600,800)

df <- data.frame(loan,score)

ggplot(df,aes(loan,score))+
geom_point()

ggplot(df,aes(y=loan))+
geom_boxplot()

ggplot(df,aes(loan))+
geom_histogram(binwidth=50000)
```


1.00
0.75
0.50
0.25
0.00
count
100000
150000
200000
250000
300000
loan